$Na_2S_2O_4$-Promoted Radical Addition Reaction of Perfluoroalkyl Iodides with Allenes and the Pd(0)-Catalyzed Stereoselective Sonogashira Coupling Reaction of Addition Products with Propargyl Alcohol

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Received 23 September 2005

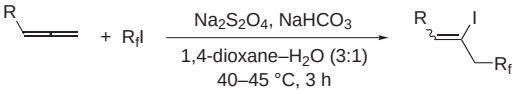
Abstract: A $Na_2S_2O_4$-promoted radical addition reaction of perfluoroalkyl iodides with allenes affording 1-perfluoroalkyl-2-alken-2-yl iodides **3** in 52–69% yields has been developed. Kinetic resolution of the *Z*/*E* mixture via the Pd(0)-catalyzed Sonogashira coupling reaction was applied to prepare pure vinylic iodides (*Z*)-**3** and the related coupling products (*E*)-**4** up to 98:2 *E*/*Z* ratio.

Key words: perfluoroalkyl iodides, allenes, radical addition, propargyl alcohol, Sonogashira coupling

For a long period of time, allenes have been considered as highly unstable, which has seriously retarded the development of their chemistry.[1] The situation has changed dramatically during the last ten years, now allenes have become an important class of compounds with unique reactivity and selectivity in organic synthesis.[2,3] In particular, the transition metal-catalyzed or -promoted reactions of allenes have become very powerful tools for organic transformations.[4] However, the radical chemistry of allenes has not been well-established.[5,6] In this paper, we wish to disclose our recent observations on the $Na_2S_2O_4$-promoted radical addition of perfluoroalkyl iodides with allenes, which afforded a *Z*/*E* mixture of 1-perfluoroalkyl-2-alken-2-yl iodides. These products may be efficiently resolved by the Pd(0)-catalyzed Sonogashira coupling reactions with propargylic alcohols.

It is well-known that $Na_2S_2O_4$ can be used to promote the radical addition of perfluoroalkyl iodides with alkenes[7] or alkynes.[8] However, when we treated 4-phenyl-1,2-butadiene with $I(CF_2)_4Cl$ in CH_2Cl_2–H_2O (3:1) or benzene–H_2O (3:1), no expected product **3a** was formed; when the reaction was carried out in aqueous DMF, DMA, DMSO, or THF, **3a** was formed in 15–25% yields. Best results were obtained when 1,4-dioxane was used as the solvent. Thus, we chose 1,4-dioxane as the solvent to study the scope of this radical addition reaction. Some of the most typical results are shown in Table 1;[11] it can be concluded that either benzyl-, *n*-alkyl-, or cyclohexyl-substituted allenes can undergo the radical addition reaction with perfluoroalkyl iodides affording the radical addition products vinylic iodides **3** in 52–69% yields.

Table 1 $Na_2S_2O_4$-Initiated Radical Addition of Perfluoroalkyl Iodides with Different Allenes[a]



1 2 3

Entry	R	R^f	Yield of **3** (%)[b]	*E*/*Z*[c]
1	Bn (**1a**)	$Cl(CF_2)_4$ (**2a**)	63 (**3a**)	1:1.80
2	Bn (**1a**)	*n*-C_4F_9 (**2b**)	65 (**3b**)	1:1.78
3	*n*-C_5H_{11} (**1b**)	$Cl(CF_2)_4$ (**2a**)	55 (**3c**)	1:2.41
4	*n*-C_5H_{11} (**1b**)	*n*-C_4F_9 (**2b**)	52 (**3d**)	1:2.33
5	*n*-C_7H_{15} (**1c**)	$Cl(CF_2)_4$ (**2a**)	63 (**3e**)	1:2.75
6	*n*-C_7H_{15} (**1c**)	*n*-C_4F_9 (**2b**)	69 (**3f**)	1:2.34
7	*c*-C_6H_{11} (**1d**)	$Cl(CF_2)_4$ (**2a**)	59 (**3g**)	1:2.22
8	*c*-C_6H_{11} (**1d**)	*n*-C_4F_9 (**2b**)	62 (**3h**)	1:2.21

[a] Ratio of **1**:**2**:$Na_2S_2O_4$:$NaHCO_3$ = 1:1.5:1.5:1.5.
[b] Isolated yield. The *E* and *Z* isomers cannot be separated by flash chromatography.
[c] Determined by NMR spectra.

Although the reactions afforded the products **3** as an *E*/*Z* mixture with the ratio ranging from 1:1.78 to 1:2.75, it was observed that the *Z* and *E* isomers of **3b** reacted very differently with propargyl alcohol, i.e., the *E* isomer reacted very fast producing **4b** with a *Z*/*E* ratio as high as 4:96 while (*Z*)-**3b** remained unreacted (Table 2).[9,10]

In this way (*Z*)-**3** may be prepared highly stereoselectively. After some optimization, a standard procedure was established for applying the Pd(0)-catalyzed Sonogashira coupling reaction[12] to the highly stereoselective synthesis of (*Z*)-**3** and (*E*)-**4** (Table 3).[11] From the results shown in Table 3 it can be concluded that (*Z*)-**3** can be prepared in the pure form in very high yields, while the *E*/*Z* ratio for

SYNLETT 2006, No. 8, pp 1263–1265
Advanced online publication: 10.03.2006
DOI: 10.1055/s-2006-932476; Art ID: W30205ST
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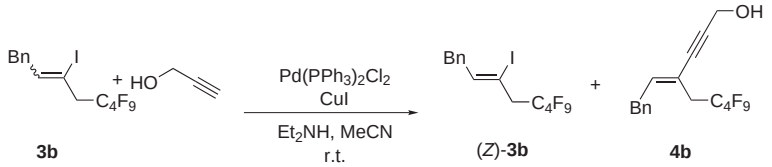
Table 2 Optimization of Reaction Conditions for the Pd(0)-Catalyzed Kinetic Resolution Reaction of **3b** with Propargyl Alcohol



3b (*Z*)-**3b** **4b**

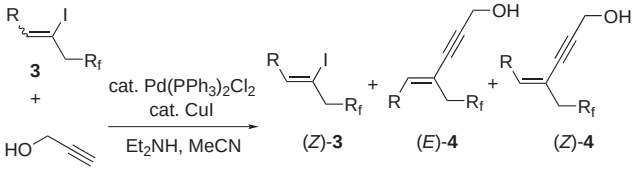
Entry	Alcohol–Et_2NH, 1:1 (equiv)	Catalyst (mol%)	Time (min)	Yield of (*Z*)-**3b** (%)	Yield of **4b** (%)	*E*/*Z* ratio of **4b**[a]
1	0.6	3.0	50	54[b]	26	95:5
2	0.7	3.5	25	48	29	93:7
3	0.9	4.5	12	61	33	96:4
4	0.6	4.5	30	54	30	96:4
5	**0.6**	**3.5**	**30**	**57**	**32**	**95:5**
6	0.7	3.0	12	53	34	91:9
7	0.9	3.0	12	50	35	88:12

[a] The *Z*/*E* ratio of **4b** was determined by GC analysis.
[b] 16% of (*E*)-**3b** remained unreacted.

the Sonogashira coupling products **4** ranging from 9:1 to 22:1. These two isomers may also be easily separated via chromatography on silica gel, enabling the synthesis of both (*Z*)-**4** and (*E*)-**4**.

In conclusion, we have developed a $Na_2S_2O_4$-promoted radical addition of perfluoroalkyl iodides with allenes. Although the stereoselectivity is not good and the *Z*/*E* isomers are inseparable, the subsequent Pd(0)-catalyzed Sonogashira coupling reaction enables the highly stereoselective synthesis of the Sonogashira coupling products (*E*)-4-perfluoroalkylmethyl-4-alken-2-ynol [(*E*)-**4**] and the *Z* isomer of the radical addition products (*Z*)-**3**. Further studies in this area are being carried out in our laboratory.

Table 3 Kinetic Resolution of Different 1-Perfluoroalkyl-2-iodo-2-alkenes by the Pd(0)-Catalyzed Sonogashira Coupling Reaction with Propargyl Alcohol[a,b]



(*Z*)-**3** (*E*)-**4** (*Z*)-**4**

R	R_f	Temp (°C)/Time (min)	Yield of (*Z*)-**3** (%)[a]	Yield of (*E*)-**4** (%)[a]	Yield of (*Z*)-**4** (%)[a]
Bn	$Cl(CF_2)_4$ (**3a**)	11/30	55	28	3
n-C_5H_{11}	C_4F_8Cl (**3c**)	0/60	45	22	2
n-C_7H_{15}	*n*-C_4F_9 (**3f**)	0/60	58	22	1
c-C_6H_{11}	$Cl(CF_2)_4$ (**3g**)	13/30	47	25	2

[a] Isolated yield.
[b] Reaction conditions: **3**:propargyl alcohol:$Pd(PPh_3)_2Cl_2$:CuI = 1:0.6:0.035:0.035 (molar ratio).

Acknowledgment

Financial support from the National Natural Science Foundation of China, the Major State Basic Research Development Program (Grant No. G2000077500), and Cheung Kong Scholar Program is greatly appreciated. Shengming Ma is jointly appointed by Zhejiang University and Shanghai Institute of Organic Chemistry. This work was conducted at Zhejiang University.

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(11) **Typical Procedure.**
The reaction of **1a** (258.8 mg, 1.99 mmol), IC_4F_9 (**2b**, 1.0482 g, 3.03 mmol), $Na_2S_2O_4$ (523.6 mg, 3.01 mmol), and $NaHCO_3$ (253.4 mg, 3.02 mmol) in 16 mL of mixed solvent of 1,4-dioxane and H_2O (3:1) afforded 612.7 mg (65%) of **3b**. ^1H NMR (400 MHz, $CDCl_3$): δ = 7.37–7.28 (m, 2 H), 7.28–7.20 (m, 2 H), 7.20–7.14 (m, 1 H), 6.84 (*E* isomer, t, *J* = 7.6 Hz, 0.36 H), 6.04 (*Z* isomer, t, *J* = 6.6 Hz, 0.64 H), 3.61–3.37 (m, 4 H). The *E*/*Z* mixture was subjected to the next step without further characterization.
Then the reaction of **3b** (118.7 mg, 0.25 mmol, *E*/*Z* = 1:1.78), propargyl alcohol (8.5 mg, 0.15 mmol), $Pd(PPh_3)_2Cl_2$ (6.2 mg, 8.84×10^{-3} mmol, 3.5 mol%), CuI (1.8 mg, 9.47×10^{-3} mmol, 3.8 mol%), and Et_2NH (11.0 mg, 0.15 mmol) in 0.5 mL of MeCN afforded 50.9 mg (43%) of (*Z*)-**3b**, 4.0 mg (4%) of (*Z*)-**4b**, and 33.9 mg (34%) of (*E*)-**4b**.

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